

Week 2: Data Basics

Understanding Data Structures & Cleaning Fundamentals

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Abstract

This presentation covers Week 2 of the AI and Machine Learning Foundation Program, focusing on Data Basics. It is designed for non-programmers to master the foundational skills required before feeding data into Machine Learning models.

- **Core Concepts:** Understanding the fundamental role of data in AI and differentiating between structured and unstructured formats.
- **Data Handling:** Navigating datasets and Comma-Separated Values (CSV) files using standard spreadsheet tools.
- **Data Cleaning:** Mastering techniques like sorting, filtering, deduplication, and handling missing values.
- **Practical Application:** Executing a guided practical activity to clean, organize, and document a sample dataset for reproducibility.

Session Objectives

By the end of this 3-day module, learners will be able to:

- Define data and distinguish between structured and unstructured formats.
- Identify various data types (text, numerical, categorical, images).
- Open, navigate, and understand the anatomy of CSV datasets.
- Apply foundational data cleaning skills (sorting, filtering, deduplication).
- Handle missing values using appropriate imputation or deletion strategies.
- Document data cleaning processes to ensure reproducibility in ML projects.

Part 1: Understanding Data & Datasets

Day 1: Thursday – "The Foundation of AI"

What is Data?

- Data consists of facts, statistics, or information collected for reference or analysis.
- In the context of AI, data is the raw material from which models learn patterns.
- Without data, an AI model is like an engine without fuel—it cannot function or make predictions.
- The quality and quantity of data directly impact the intelligence and accuracy of the AI.

Key Takeaway

AI does not "know" things; it "learns" from examples. Data provides those examples.

Data: The Fuel of AI

- The more relevant and high-quality data you provide, the better the model performs.
- Data tells the model what patterns to look for (e.g., features of a spam email, indicators of customer churn).
- "Data is the new oil" – but like crude oil, it must be refined (cleaned) to be useful.

The Reality of AI Projects

Data collection and preparation often take up to 80% of a Data Scientist's time. The actual model training is just the final step.

Structured vs. Unstructured Data

Structured Data

- Highly organized.
- Formatted in a predefined manner (tables).
- Easily searchable and relational.
- *Example:* Bank transaction records.

Unstructured Data

- Free-form text or media.
- Lacks a predefined data model.
- Difficult to organize using traditional tools.
- *Example:* Customer support emails.

Most enterprise data (approx. 80%) is unstructured, but structured data is the primary fuel for traditional ML models.

Deep Dive: Structured Data

- **Characteristics:**
  - Fits neatly into rows and columns (tables).
  - Each column has a specific data type (e.g., date, integer, string).
  - Easily queried using SQL or spreadsheet formulas.
- **Examples:**
  - Relational databases (MySQL, PostgreSQL).
  - Spreadsheets (Excel, Google Sheets).
  - Student grades, inventory logs, sales receipts.

Deep Dive: Unstructured Data

- **Characteristics:**
  - No predefined format or organization.
  - Requires advanced NLP (Natural Language Processing) or Computer Vision to extract meaning.
  - Rich in context but computationally expensive to process.
- **Examples:**
  - Emails, social media posts, chat logs.
  - Images, audio files, video recordings.
  - PDF documents, Word files.

Types of Data in AI

Data Type	Description	AI Application
Text	Words, sentences, documents	NLP (Chatbots, Translation)
Numbers	Quantitative measurements	Statistical Analysis, Forecasting
Images	Pixels, shapes, colors	Computer Vision (Facial Recognition)
Audio	Sound waves, speech	Speech-to-Text, Voice Assistants

Focus for Week 2

In this week, we focus primarily on **Text** and **Numbers** as they form the basis of the tabular data we will clean in Excel/Sheets.

Categorical vs. Numerical Data

Numerical Data

Represents quantities.

- **Discrete:** Countable (e.g., number of children, items sold).
- **Continuous:** Measurable (e.g., height, temperature, salary).

Categorical Data

Represents characteristics or groups.

- **Nominal:** No order (e.g., gender, color, branch).
- **Ordinal:** Inherent order (e.g., satisfaction: Low, Med, High).

Distinguishing between these is crucial because ML algorithms process numbers and categories using entirely different mathematical techniques.

Interactive Discussion: Kenyan Context

Let's identify data types in everyday Kenyan businesses:

Mobile Lending Apps (e.g., M-Shwari, Fuliza)

- **Structured:** Loan amount, repayment date, M-Pesa transaction IDs.
- **Unstructured:** Customer support chat logs, voice notes from field agents.

Agriculture (e.g., DigiFarm)

- **Structured:** Crop yield per acre, fertilizer quantity purchased.
- **Unstructured:** Drone imagery of farms, localized weather reports.

Part 2: Datasets & CSV Files

"Understanding the standard format for ML data."

What is a CSV File?

- **CSV** stands for **C**omma-**S**eparated **V**alues.
- It is a plain text file that stores tabular data.
- Each line in the file represents a row in the table.
- Commas (or sometimes tabs/semicolons) separate the values in each column.
- It is the universal standard for exchanging data between different systems, databases, and ML tools.

Why CSV?

Unlike Excel files (`.xlsx`), CSVs do not contain formatting, formulas, or macros. They are pure data, making them lightweight and universally compatible.

Anatomy of a Dataset

A dataset is a structured collection of data. In tabular format, it consists of:

- **Features (Variables/Columns):** The attributes being measured or recorded (e.g., Age, Salary, Location).
- **Observations (Records/Rows):** Individual instances or data points (e.g., Customer 1, Transaction A).
- **Target Variable:** The specific feature we want the ML model to predict (e.g., "Did the customer churn? Yes/No").

Rows, Columns, and Headers

CustomerID	Age	City	Churned
101	34	Nairobi	No
102	45	Mombasa	Yes
103	28	Kisumu	No

- **Headers:** The first row, containing the names of the columns (features).
- **Columns:** Vertical arrays of data of the same type.
- **Rows:** Horizontal arrays representing a single entity or event.

Tool Setup: Excel & Google Sheets

We will use spreadsheet software to view and clean our data.

Microsoft Excel

- Powerful offline tool.
- Great for large datasets.
- Advanced formulas and Power Query.

Google Sheets

- Browser-based.
- Excellent for collaboration.
- Cloud storage and easy sharing.

Both tools can easily open, edit, and export CSV files. Ensure you have access to at least one before proceeding.

Hands-on Mini-Task

1. **Download** the sample raw CSV file provided by the instructor.
2. **Open** the file in Google Sheets or Excel.
   - *Troubleshooting:* If it opens as a single column of text, use the "Text to Columns" feature (`Data > Text to Columns`) and select "Comma" as the delimiter.
3. **Observe** how the raw text transforms into a structured, readable table.

Instructor Check

Raise your hand if your data is now properly separated into columns and rows.

Identifying Data Types in Sheets

Look at your opened CSV and identify the data types:

- **Numbers:** Which columns contain pure numbers? (e.g., Age, Price, Quantity)
- **Text/Categorical:** Which columns contain words or categories? (e.g., Name, City, Status)
- **Dates:** Are there any dates? (e.g., Join Date, Transaction Time)

Crucial Step

Spreadsheets automatically guess data types, but as Data Scientists, we must verify them. If a "Number" column contains a single text character (like a space), the ML model will fail to process it.

Day 1 Assignment

Task

- Find **one example** of structured data and **one example** of unstructured data in your daily life or workplace.
- Write a short paragraph (3-4 sentences) on how an AI might use *both* to solve a specific business problem.

Submission

Upload your short paragraph to the Mediacrest learning portal by tomorrow morning.

Part 3: Data Cleaning Fundamentals

Day 2: Friday – "Garbage In, Garbage Out"

The Golden Rule: Garbage In, Garbage Out

- Machine Learning models are highly sensitive to the quality of data.
- If you train a model on messy, inaccurate, or biased data, the predictions will be fundamentally flawed.
- Data cleaning is not just "admin work"; it is the most critical step in the ML pipeline.
- It ensures the model learns true patterns, not artifacts of poor data collection.

The Mantra

"Clean data is the foundation of powerful AI."

Sorting and Filtering

Sorting

Arranging data in a meaningful order (e.g., A-Z, lowest to highest).

- Helps in quickly spotting outliers.
- Makes trends visually obvious.

Filtering

Hiding rows that do not meet specific criteria.

- Allows focus on a subset (e.g., only "Active" users).
- Essential for Exploratory Data Analysis (EDA).

Both are essential first steps to understand the "shape" and distribution of your dataset before applying complex cleaning rules.

Identifying and Removing Duplicates

- **Why it happens:** Double data entry, system glitches, or merging multiple data sources.
- **Impact on ML:** Models might over-weight duplicated records, leading to biased predictions and overfitting.

How to remove in Sheets/Excel

1. Select the entire data range.
2. Go to `Data > Data cleanup > Remove duplicates` (Sheets) or `Data > Remove Duplicates` (Excel).
3. **Best Practice:** Always choose "Select all columns" to ensure entire row uniqueness based on a primary key (like Customer ID).

The Challenge of Missing Values

- **How it appears:** Blank cells, "N/A", "Null", "NA", "Unknown", or "-".
- **Why it happens:** User error, system failures, or the question was not applicable to the user.
- **Why it's a problem:** Most ML algorithms cannot process blank cells. They will either throw an error and crash, or silently ignore the data, reducing model accuracy.

The Rule

We must address missing values *before* feeding the data into any Machine Learning algorithm.

Strategies for Missing Data

Deletion (Dropping)

- Remove the **row** if only a tiny fraction of data is missing.
- Remove the **column** if it is mostly empty (e.g., $> 50\%$ missing).

Imputation (Filling)

- Fill with **Mean** (average) for numerical data.
- Fill with **Median** for skewed data.
- Fill with **Mode** (most frequent) for categories.
- Fill with a placeholder (e.g., "Unknown").

The choice of strategy depends on *why* the data is missing and the volume of missing records.

Hands-on Exercise

Open the "messy" sample dataset provided for Day 2.

1. **Sort** the "Sales" column from highest to lowest.
2. **Filter** to show only "Active" customers.
3. **Remove** any duplicate Customer IDs.
4. **Identify** missing values in the "Age" column.
5. **Impute** the missing "Age" values using the Median age of the dataset.

Instructor Check

Save your file as `dataset_cleaned_day2.csv` and keep it ready for tomorrow's documentation session.

Day 2 Assignment

Task

- Complete the "Data Cleaning Checklist" worksheet provided by the instructor.
- Note every tool/button used to clean the Friday dataset.
- Reflect on which missing value strategy you chose for the "Age" column and write a brief justification.

Submission

Upload the cleaned dataset and the completed checklist to the portal before Saturday's session.

Part 4: Practical Activity & Documentation

Day 3: Saturday – "The Data Cleaning Pipeline"

Why Documentation Matters in AI/ML

- **Reproducibility:** Other data scientists (or you in 6 months) must be able to recreate your exact steps to achieve the same results.
- **Transparency:** Business stakeholders need to know how data was altered before making critical decisions based on your model.
- **Auditability:** In regulated industries, you must prove data was handled ethically and correctly.

The Reality

A model is only as good as the documented pipeline that created it. Undocumented cleaning steps are a massive risk in AI deployment.

The Data Dictionary & Cleaning Log

Data Dictionary

A document describing the dataset.

- Column names.
- Data types.
- Descriptions.
- Allowed values.

Cleaning Log

A step-by-step record of modifications.

- *Step 1:* Removed 15 duplicate rows.
- *Step 2:* Imputed 5 missing 'Age' values with median (34).

This log becomes the foundation of your ML project report and ensures your work can be audited.

Best Practices in Data Handling

1. **Never overwrite the original raw data!**
2. Always keep a master copy of the raw CSV (e.g., `data_raw.csv`).
3. Save your cleaned version as a new file (e.g., `data_cleaned_v1.csv`).
4. Use version control or clear naming conventions (`v1`, `v2`) to track changes.
5. This ensures you can always revert if a cleaning step introduces errors.

Golden Rule

Treat your raw data like a museum artifact. You can study it, copy it, and clean the copy, but never alter the original.

Guided Practical Activity (Capstone)

The Task: Clean a provided "Student Performance" or "Customer Churn" raw CSV.

1. **Explore:** Check dimensions and spot obvious errors.
2. **Clean:** Remove duplicates, handle missing values, fix formatting.
3. **Organize:** Sort and filter logically.
4. **Document:** Write the Cleaning Log in a separate sheet or Word doc.
5. **Export:** Save as `dataset_cleaned_final.csv`.

Peer Review

Swap with a partner. Review each other's logs. Could your partner reproduce your exact cleaned dataset using only your log?

Week 3 Preview

Next Week: Principles of Machine Learning

- We will transition from handling data to teaching machines how to learn from it.
- Topics include:
  - Supervised vs. Unsupervised Learning.
  - Classification vs. Regression.
  - Training data vs. Testing data.
  - How ML models conceptually "learn".

Preparation

Bring your cleaned datasets from today's practical! We will use them to understand how data is split for model training.

Q&A and Thank You

Questions?

"Data without cleaning is just noise."

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